



ALLEN INSTITUTE *for*
NEURAL DYNAMICS

USING THE ALLEN BRAIN OBSERVATORY TO EXPLORE NEURAL CODING

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About me

- Bench scientist. Molecular biology -> neurophysiology
- No formal CS training. “You can pick it up if you need it”
 - (This is also true for statistics. This was bad advice.)
- Learned how to code by using code of postdoc in my lab and modifying it for my purposes.
 - Igor Pro -> Matlab -> C++ -> Python
- Joined Allen Institute in 2012
- Co-lead of the Allen Brain Observatory Visual Coding 2-photon pipeline
- Currently the Associate Director, Data and Outreach for Allen Institute for Neural Dynamics

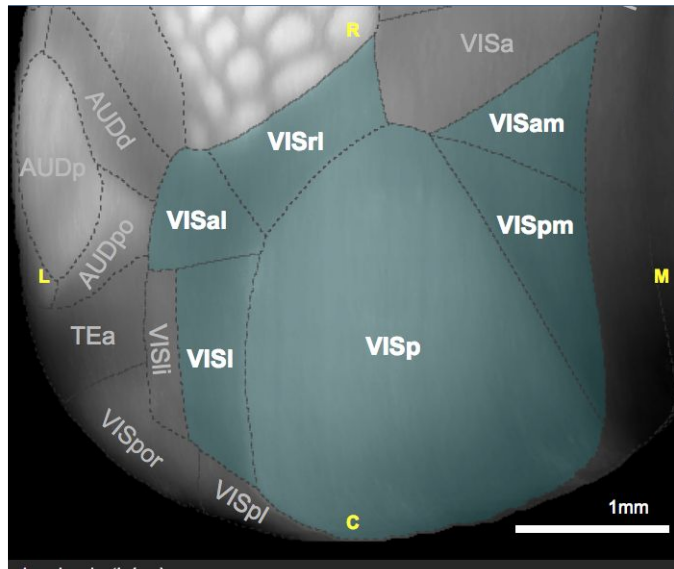
Intro to lesson

The Allen Brain Observatory: a physiological survey of the mouse visual cortex

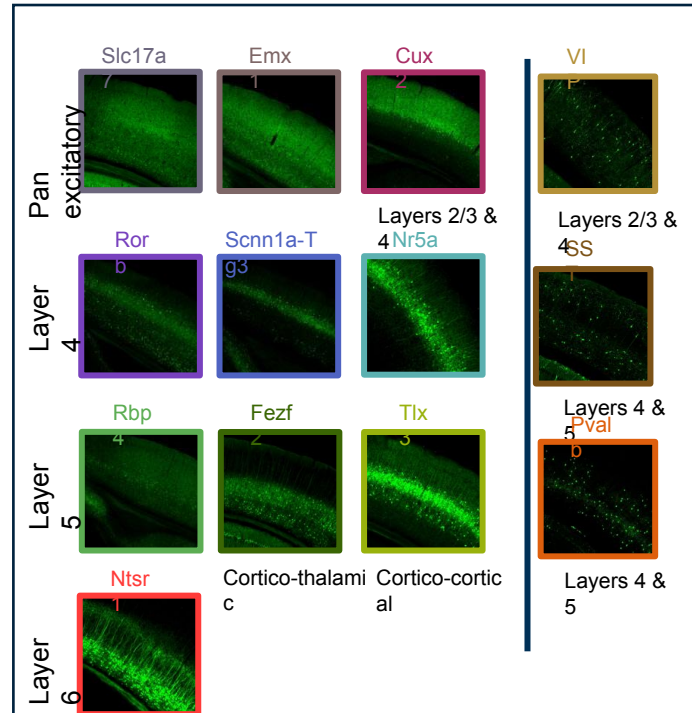
Cortical Visual Areas

Cortical Layers
Cre Lines

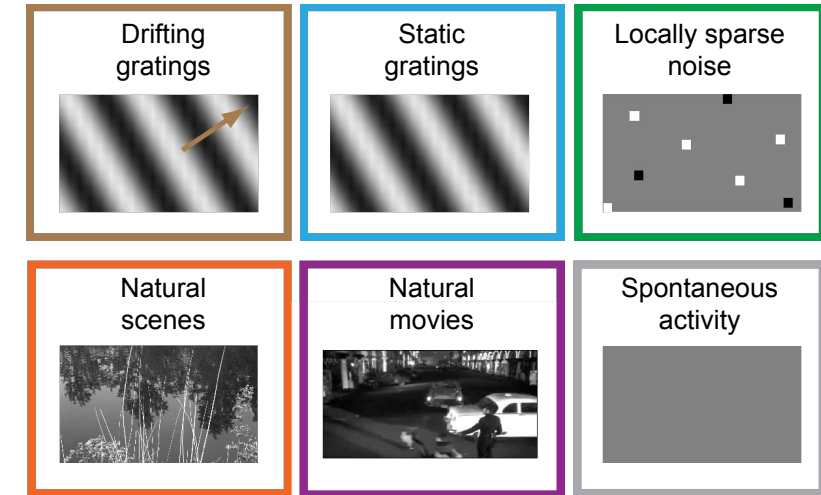
Visual Stimuli



X



X



How is visual information represented and transformed through the visual circuit?

Are there differences across cortical areas, layers and cell types?

How do stimulus statistics affect the visual encoding properties of neurons?

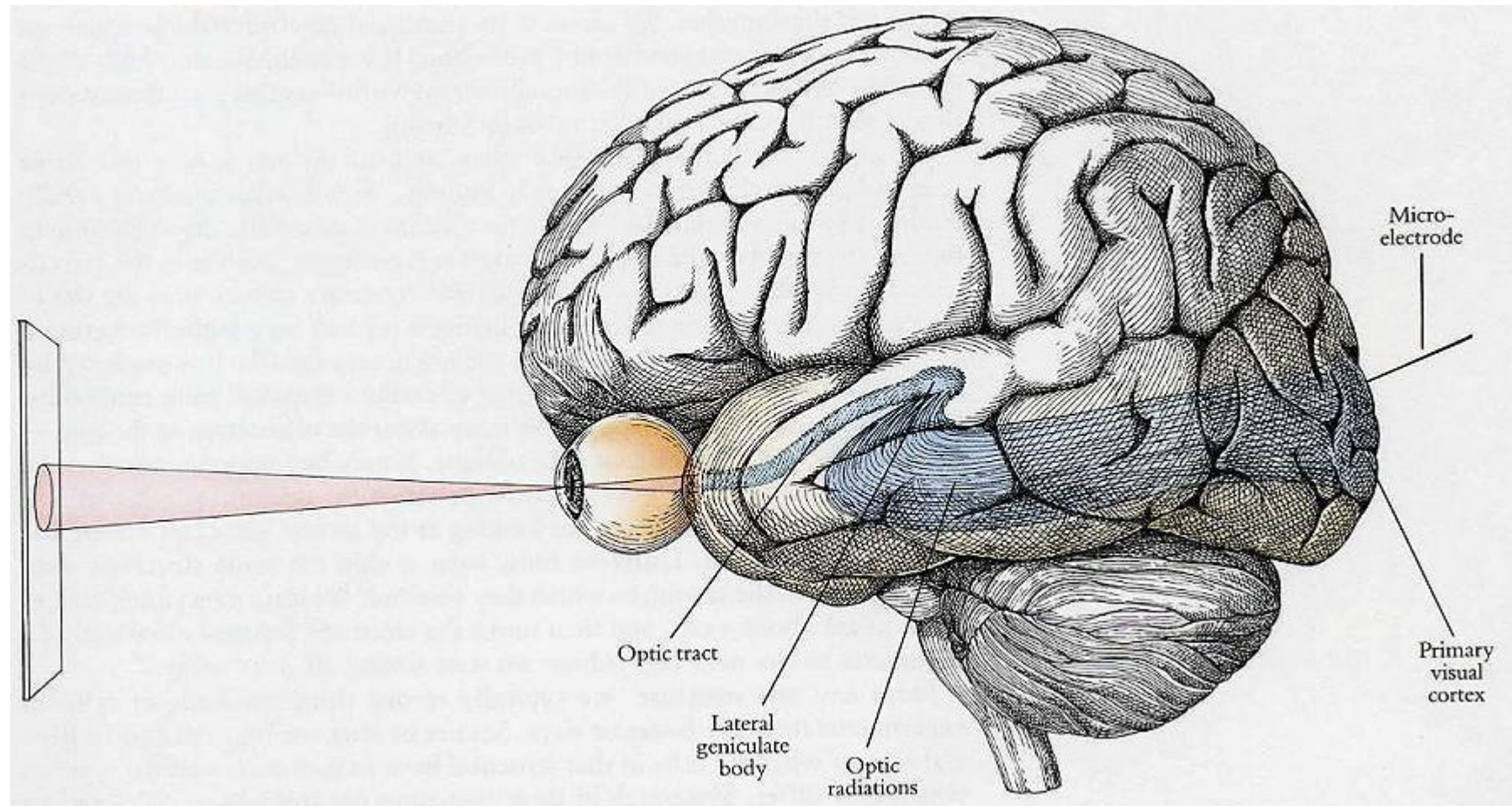
Lesson:

- Learn how to integrate neural activity and stimulus information from NWB files
- Compare tuning properties of different neurons of the same type. What kind of variability do you see?
- Compare tuning properties across cell types. How might different cell types work together to encode visual information?

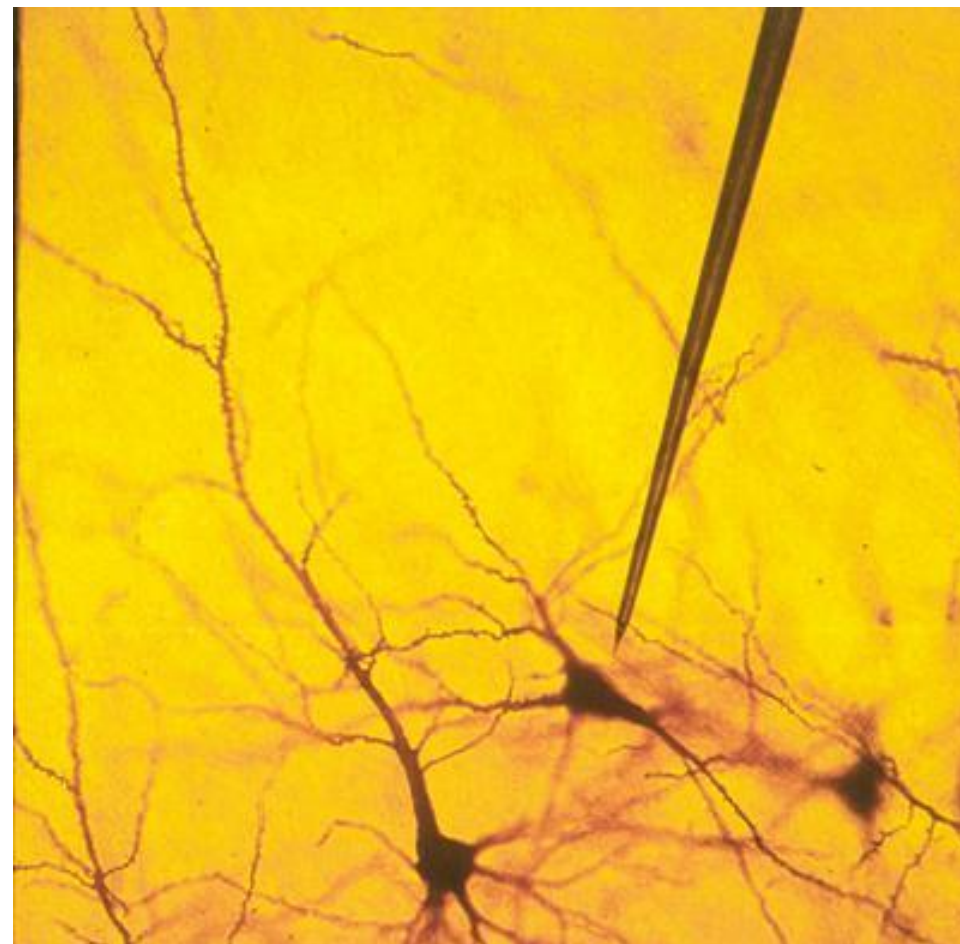
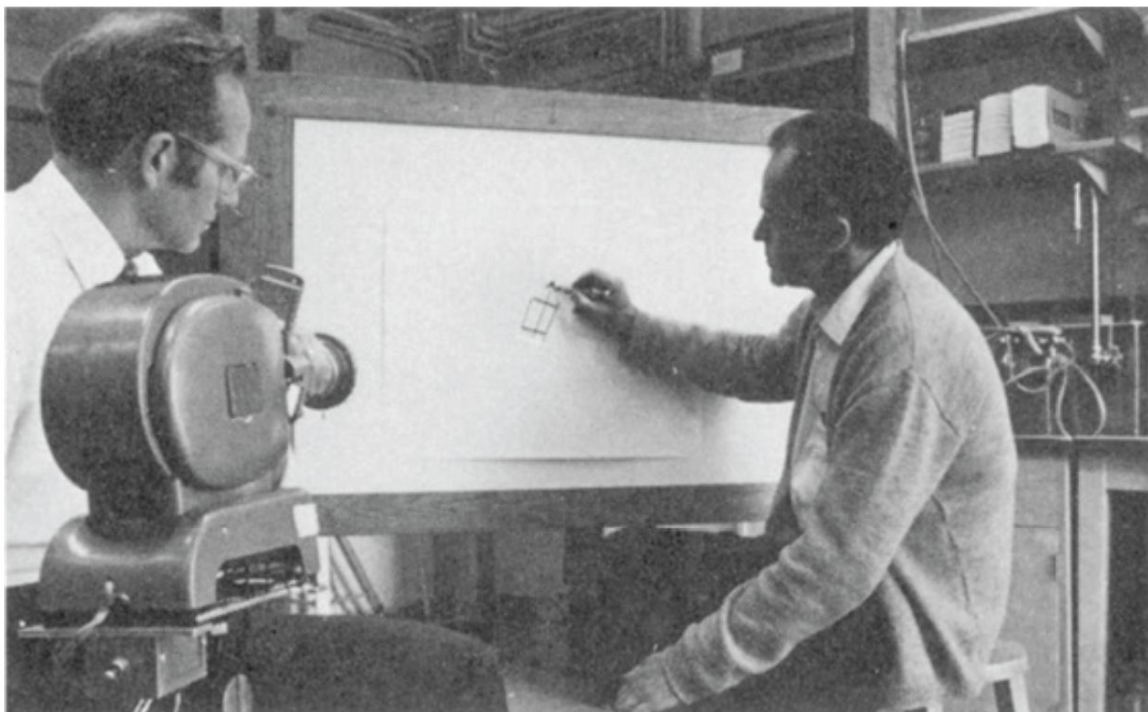
Underlying goals

- Analyze neurophysiology data
 - Questions about relation between neural activity and sensory input and/or behavior
- Use real data.
 - It can be messy!
 - It can be exciting!
- Highlight open questions in the field

Intro to Neural Coding



Hubel (1988)



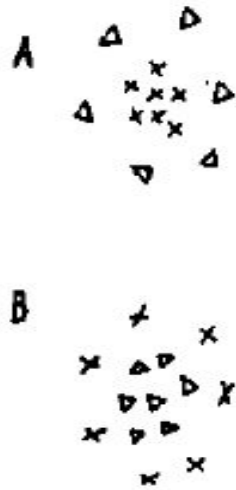
Hubel (1988)



Hubel & Wiesel (1961)

Receptive field: “area that must be illuminated to cause a cell to discharge”

LGN



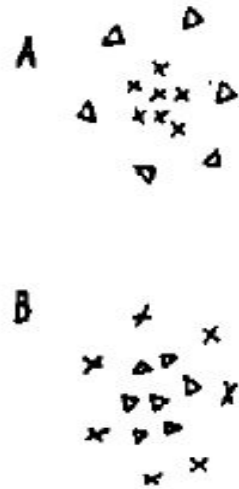
Hubel & Wiesel, 1962



*Hubel & Wiesel
(1961)*

Receptive field: “area that must be illuminated to cause a cell to discharge”

LGN

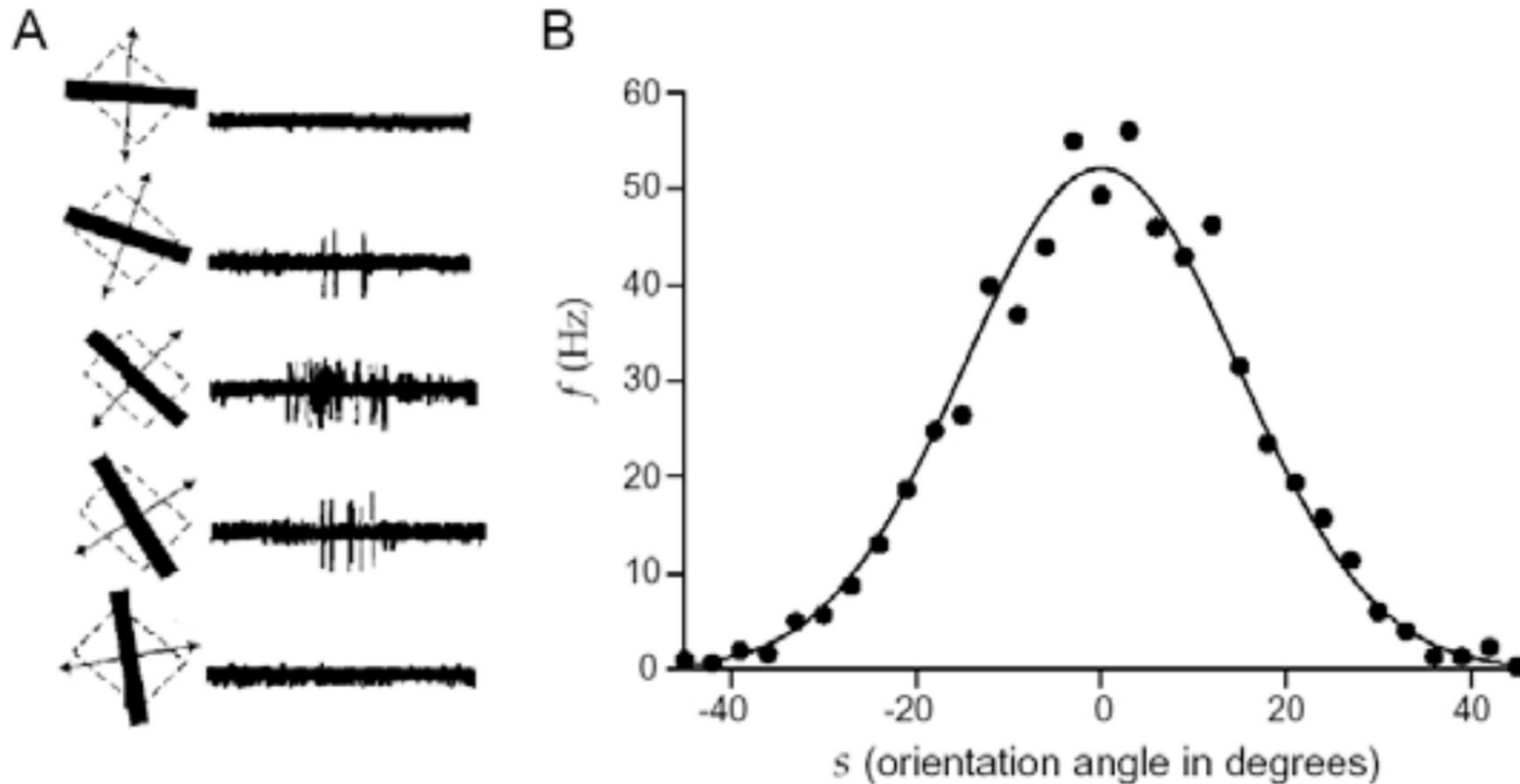


Cortex



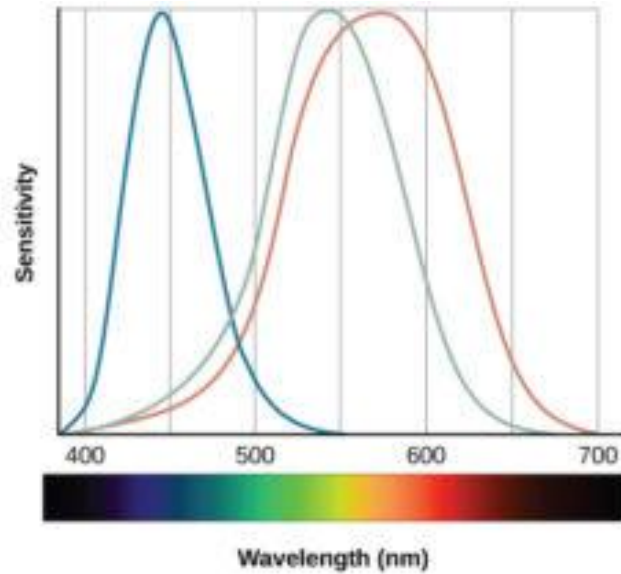
Hubel & Wiesel, 1962

Characterize orientation preference with a Tuning Curve

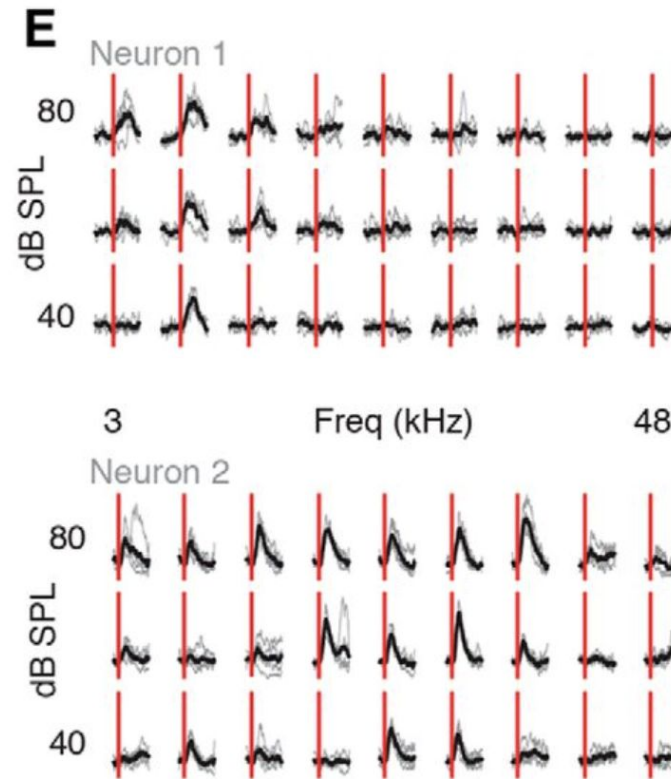


(other types of tuning curves)

Photoreceptor spectra

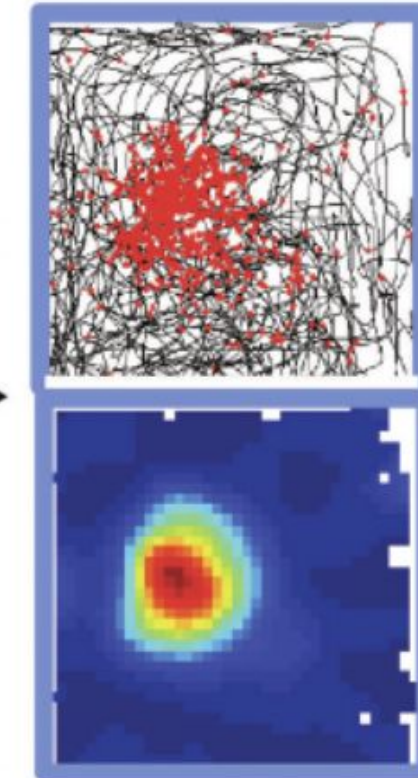


Auditory Cortex

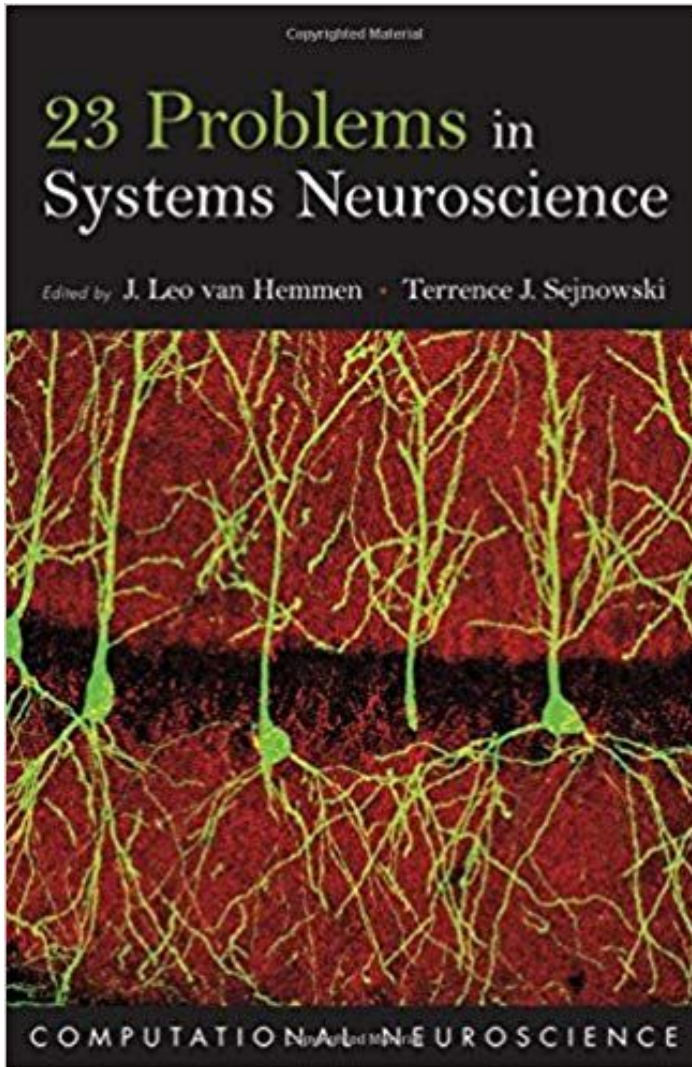


Meng *et al.* 2017

Place cells



Moser *et al.* 2015



In: Problems in Systems Neuroscience. T.J. Sejnowski, L. van Hemmen, eds. Oxford University Press. (in press)

What is the other 85% of V1 doing?

Bruno A. Olshausen

Redwood Neuroscience Institute
and Center for Neuroscience, UC Davis

David J. Field

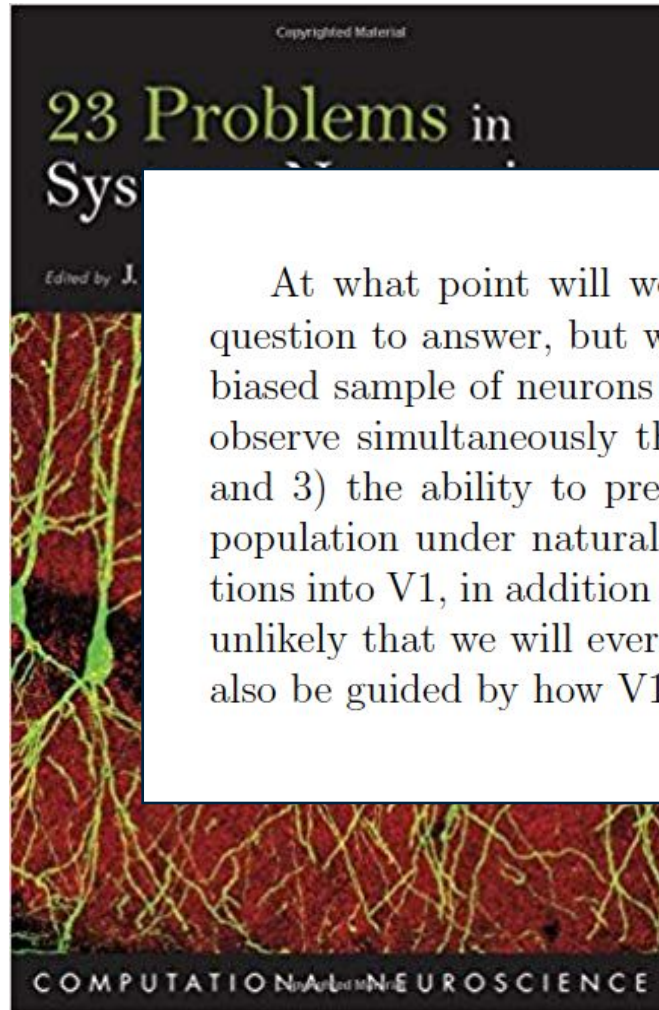
Department of Psychology
Cornell University

July 22, 2004

Abstract

This article will pose the following challenge: that despite four decades of research characterizing the response properties of V1 neurons, we still do not have a decent picture of how V1 really operates—i.e., how a population of its neurons represents natural scenes under realistic viewing conditions. We identify five problems with the current view that stem largely from biases in the design and execution of experiments, in addition to the contributions of non-linearities in the cortex that are not well understood. Our purpose is to open the window to new theories, a number of which we describe along with some proposals for testing them.

see also Carandini *et al.* 2005; Masland & Martin 2007



In: Problems in Systems Neuroscience. T.J. Sejnowski, L. van Hemmen, eds. Oxford University Press. (in press)

At what point will we actually “understand” V1? This is obviously a difficult question to answer, but we believe at least three ingredients are required: 1) an unbiased sample of neurons of all types, firing rates, and layers of V1, 2) the ability to observe simultaneously the activities of hundreds of neurons in a local population, and 3) the ability to predict, or at least qualitatively model, the responses of the population under natural viewing conditions. Given the extensive feedback connections into V1, in addition to the projections from pulvinar and other sources, it seems unlikely that we will ever understand V1 in isolation. Thus, our investigations must also be guided by how V1 fits into the bigger picture of thalamo-cortical function.

its neurons represents natural scenes under realistic viewing conditions. We identify five problems with the current view that stem largely from biases in the design and execution of experiments, in addition to the contributions of non-linearities in the cortex that are not well understood. Our purpose is to open the window to new theories, a number of which we describe along with some proposals for testing them.

Allen Brain Observatory: Visual Coding dataset

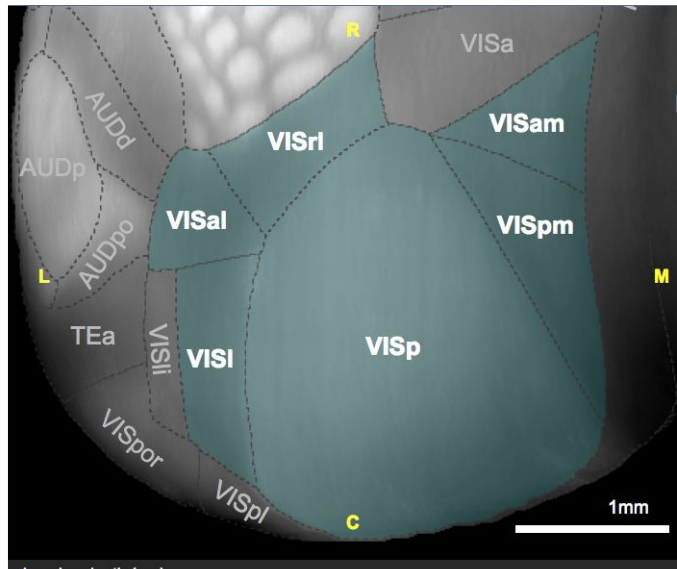
Cortical Visual Areas



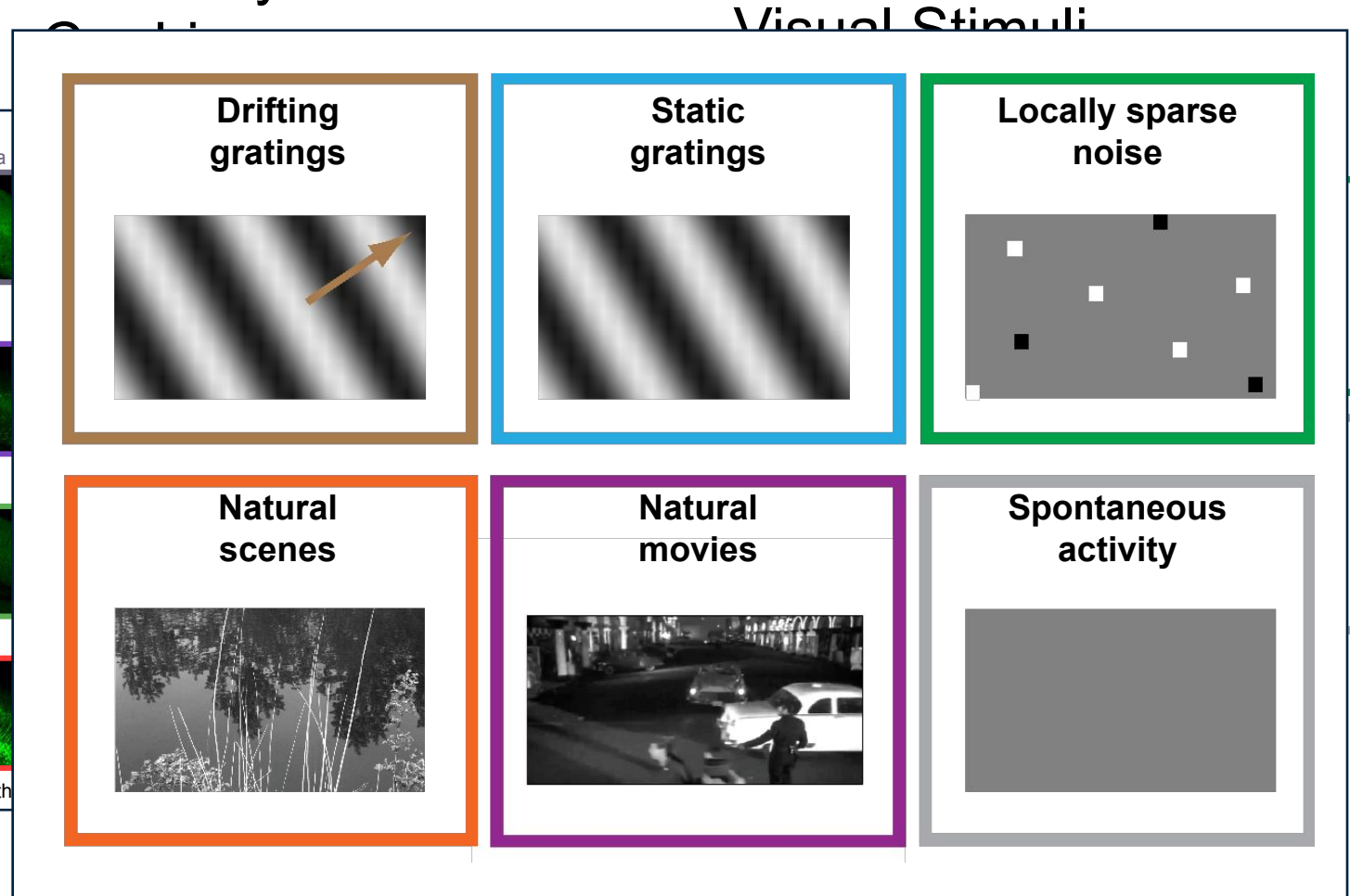
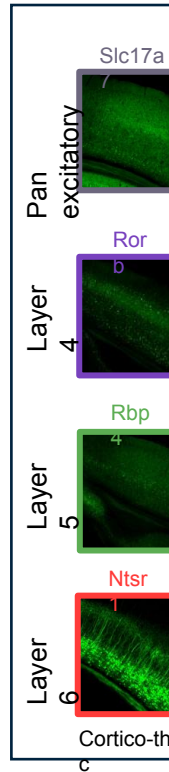
The Allen Brain Observatory: a physiological survey of the mouse visual cortex

Cortical Layers

Cortical Visual Areas



X

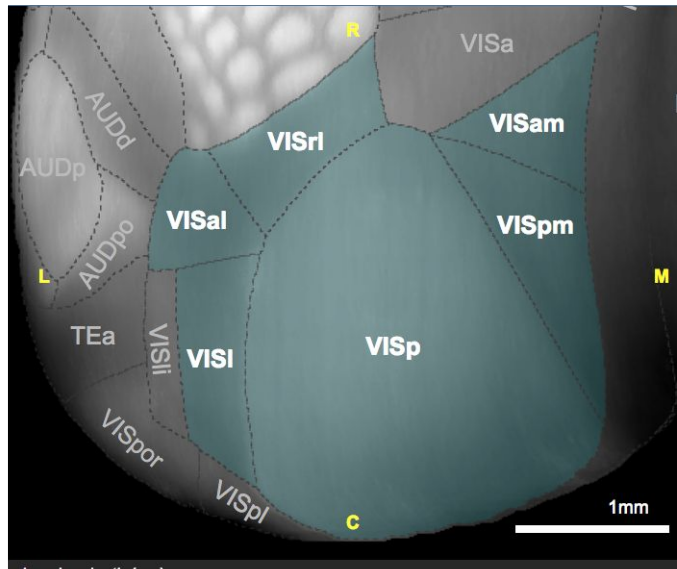


The Allen Brain Observatory: a physiological survey of the mouse visual cortex

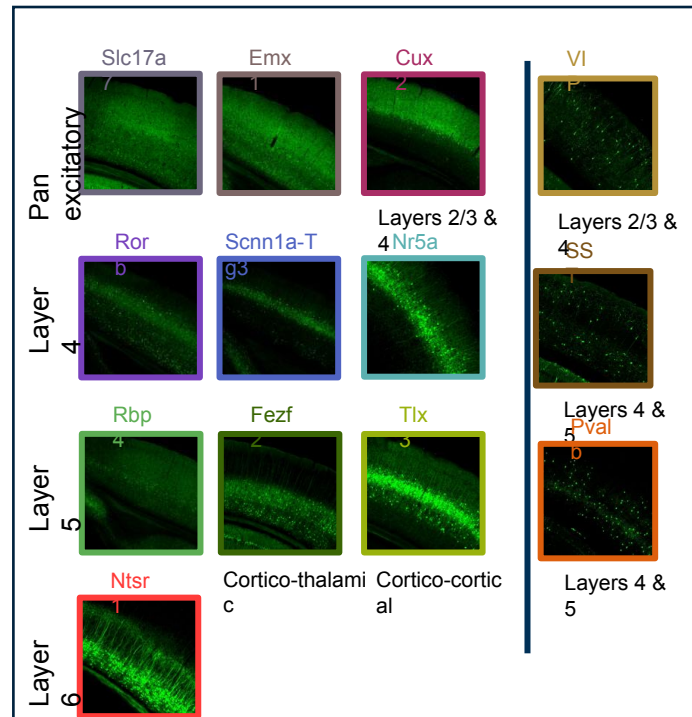
Cortical Visual Areas

Cortical Layers
Cre Lines

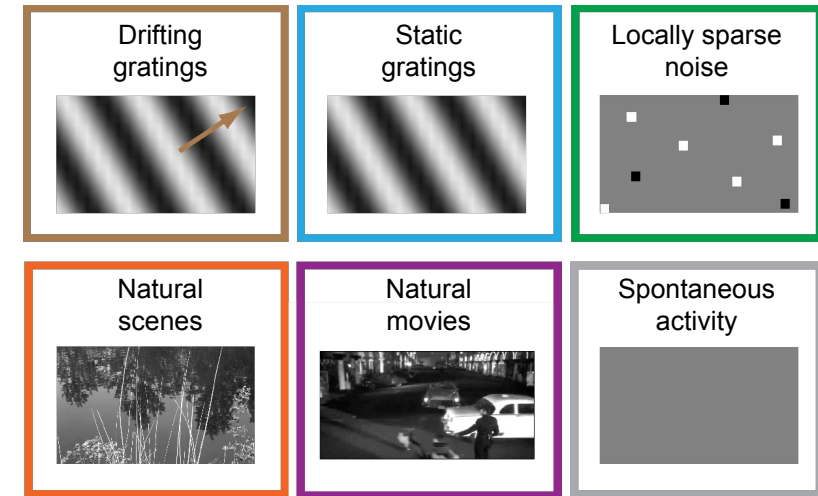
Visual Stimuli



X



X

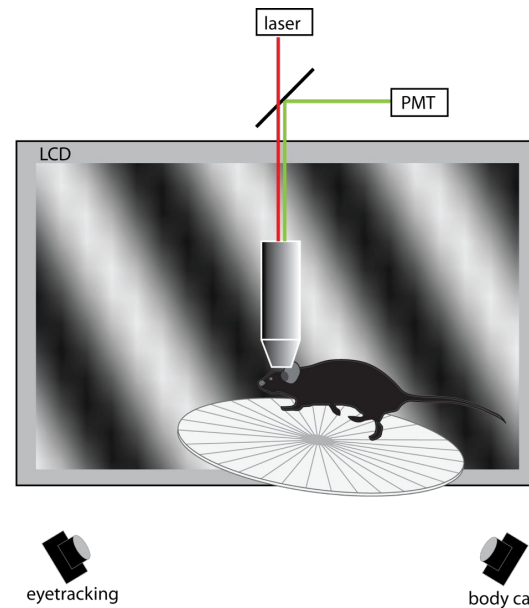
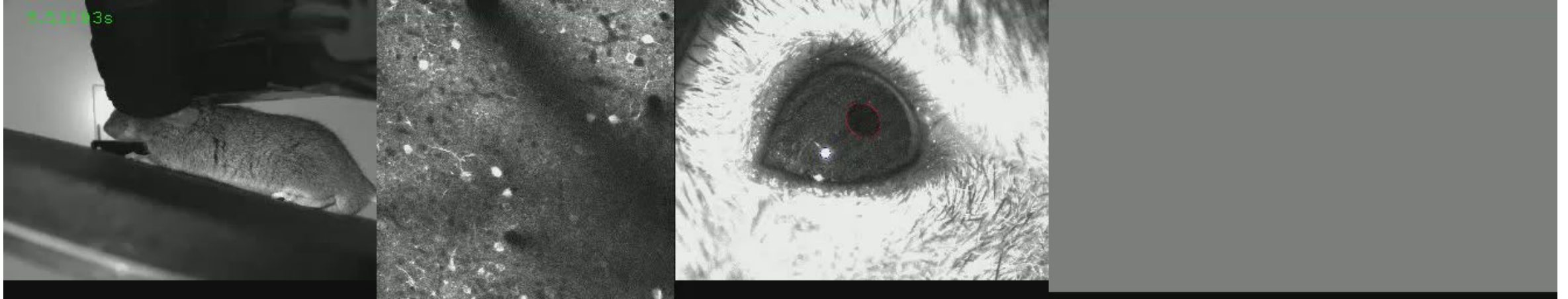


How is visual information represented and transformed through the visual circuit?

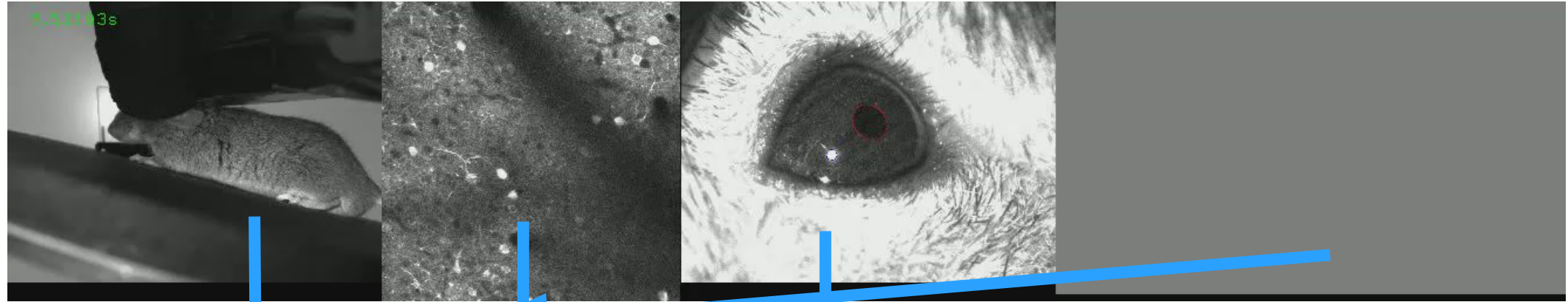
Are there differences across cortical areas, layers and cell types?

How do stimulus statistics affect the visual encoding properties of neurons?

2-photon calcium imaging in an awake mouse



Data processed and packaged in NWB file

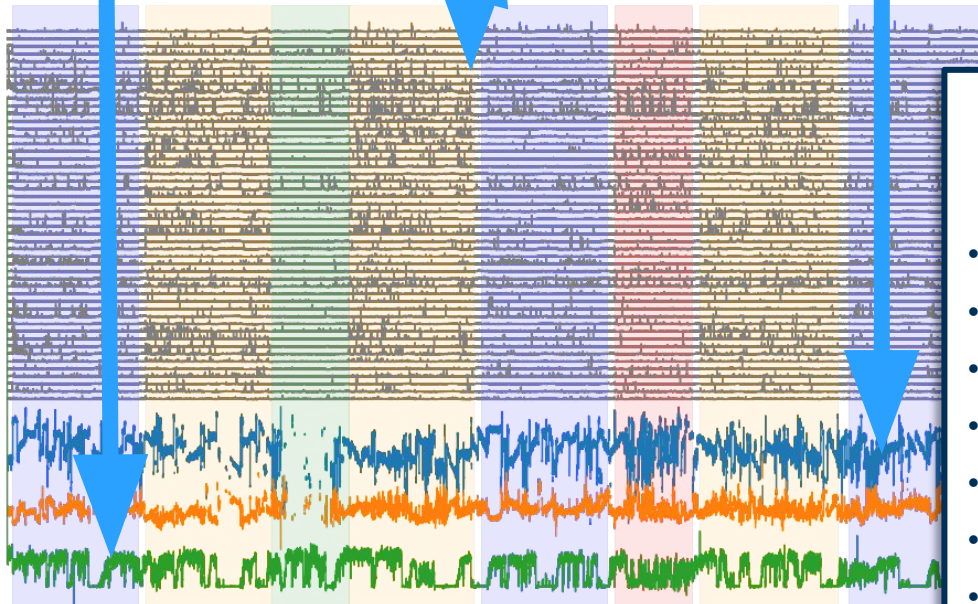


Stimulus presentation

Cell activity

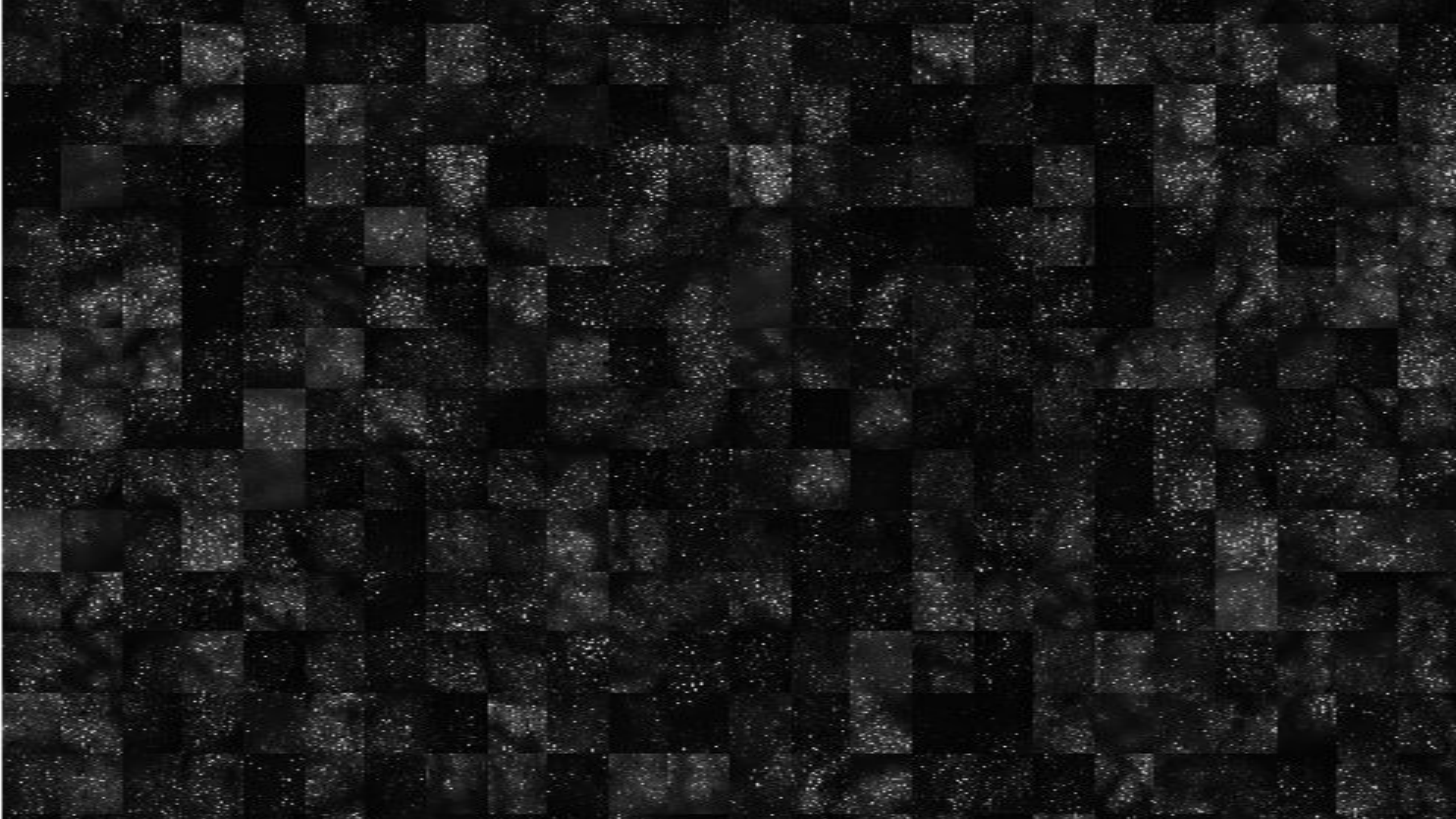
Eye position

Running speed



NEURODATA
WITHOUT BORDERS

- Max projection image
- ROI masks
- Cell specimen IDs
- Raw fluorescence for each ROI
- Surrounding neuropil trace and r value
- Global $\Delta F/F$ trace for each ROI
- Stimulus table



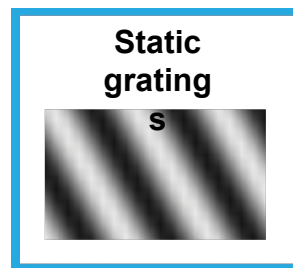
Allen Brain Observatory 2P Visual Coding Dataset

Cre line	Layers	E/I	F/S	VISp	VISI	VISal	VISpm	VISam	VISrl	Total
Emx1-IRES-Cre; Camk2a-tTA; Ai93	2/3,4,5	E	F	3073 (10)	2098 (8)	1787 (7)	835 (4)	457 (3)	3011 (9)	11261 (41)
Slc17a7-IRES2-Cre; Camk2a-tTA;Ai93	2/3,4,5	E	F	4840 (17)	3230 (16)	374 (2)	1970 (15)	235 (2)	137 (2)	10786 (54)
Slc17a7-IRES2-Cre; Camk2a-tTA;Ai94	2/3,4,5	E	S	1419 (3)						1419 (3)
Cux2-CreERT2; Camk2-tTA; Ai93	2/3, 4	E	F	5081 (16)	2792 (11)	3103 (13)	2361 (13)	1616 (11)	1829 (12)	16782 (76)
Rorb-IRES2-Cre; Camk2a-tTA; Ai93	4	E	F	2218 (8)	1191 (6)	1242 (6)	764 (7)	735 (8)	1464 (5)	7614 (40)
Scnn1a-Tg3-Cre; Camk2a-tTA; Ai93	4	E	F	1873 (9)						1873 (9)
Nr5a1-Cre; Camk2a-tTA; Ai93	4	E	F	578 (8)	421 (6)	220 (6)	331 (7)	171 (6)	1658 (6)	3379 (39)
Rbp4-Cre_KL100; Camk2a-tTA; Ai93	5	E	F	458 (7)	485 (7)	441 (6)	509 (6)	355 (8)	102 (4)	2350 (38)
Fezf2-CreER;Ai148	5	E	F	407 (4)	981 (5)					1388 (9)
Tlx3-Cre_PL56;Ai148	5	E	F	1181 (6)	946 (3)					2127 (9)
Ntsr1-Cre_GN220;Ai148	6	E	F	573 (6)	719 (7)		581 (5)			1873 (18)
Sst-IRES-Cre;Ai148	4, 5	I	F	266 (17)	301 (15)	24 (1)	247 (14)		46 (2)	884 (49)
Vip-IRES-Cre;Ai148	2/3, 4	I	F	352 (17)	315 (17)		387 (16)			1054 (50)
Pvalb-IRES-Cre;Ai162	4,5	I	S	322 (16)	139 (5)					461 (21)
Total				22641 (144)	13618 (106)	7191 (41)	7985 (87)	3569 (38)	8247 (40)	63251 (456)

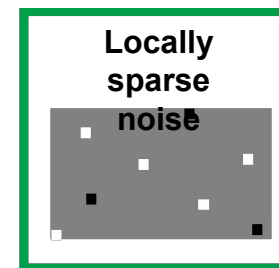
Visual Stimuli



8 directions
5 temporal frequencies
Spatial frequency = 0.04 cpd
Contrast = 80%
2 sec grating, 1 sec gray
15 trials



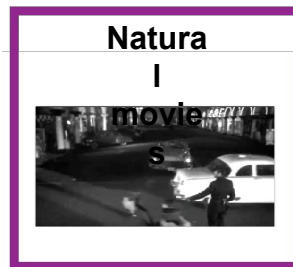
6 orientations
5 spatial frequencies
Contrast = 80%
0.25 sec presentation
50 trials



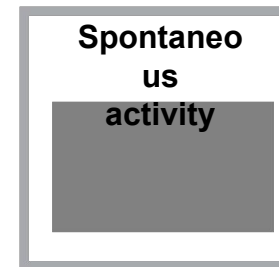
16 x 28 array of 4° & 8° squares
20° exclusion zone
0.25 sec presentation
~115 trials



118 images
0.25 sec presentation
50 trials



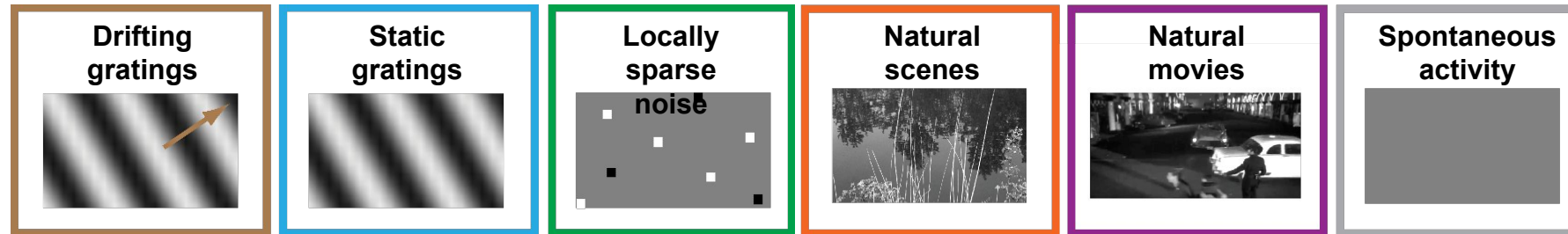
Two 30s clips
One 120s clip
10 trials



50 cd/m²
5 minutes

Stimuli are shown over 3 experimental sessions

The three **experiment sessions** for each imaging location make up a single **experiment container**



- Max projection image
- ROI masks
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- Stimulus table

Experiment Session A



- Max projection image
- ROI masks
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- Global $\Delta F/F$ trace for each ROI
- Stimulus table

Experiment Session B



- Max projection image
- ROI masks
- Cell specimen IDs
- Raw fluorescence for each ROI
- Surrounding neuropil trace and r value
- Global $\Delta F/F$ trace for each ROI
- Stimulus table

Experiment Session C

- Stimulus template (images/movies)
- Mouse's running speed
- Pupil position and area
- Motion correction (x & y pixels)
- Meta data (cre line, visual area, imaging depth, experiment container, mouse ID, age, sex, etc)

- Stimulus template (images/movies)
- Mouse's running speed
- Pupil position and area
- Motion correction (x & y pixels)
- Meta data (cre line, visual area, imaging depth, experiment container, mouse ID, age, sex, etc)

- Stimulus template (images/movies)
- Mouse's running speed
- Pupil position and area
- Motion correction (x & y pixels)
- Meta data (cre line, visual area, imaging depth, experiment container, mouse ID, age, sex, etc)

hour

Allen SDK: Python Analysis Toolkit

Load an experiment

Let's pick a random experiment from the table and plot example ophys and behavioral data.

```
# select first experiment from the table to look at in more detail.
# Note that python enumeration starts at 0.
ophys_experiment_id = selected_experiment_table.index[0]
dataset = bc.get_behavior_ophys_experiment(ophys_experiment_id)
```

```
behavior_ophys_experiment_957759562.nwb: 100% |██████████| 263M/263M [00:24<00:00, 10.7MMB/s]
```

show metadata for this experiment

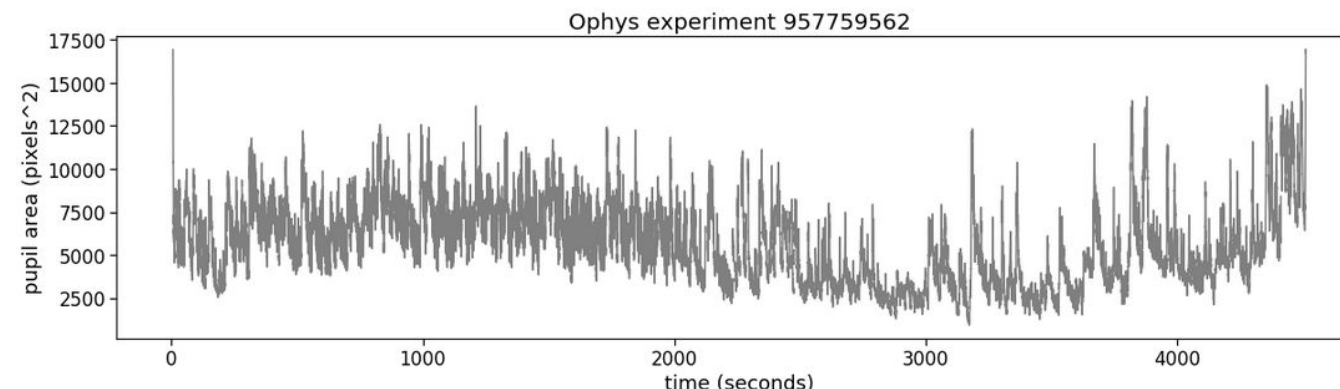
```
dataset.metadata
```

```
{'equipment_name': 'MESO.1',
 'imaging_plane_group_count': 1,
 'session_type': 'OPHYS_4 imaging',
 'ophys_experiment_id': 957759562,
 'ophys_session_id': 95702035,
 'field_of_view_width': 512,
 'behavior_session_id': 95703,
 'stimulus_frame_rate': 60.0,
 'behavior_session_uid': 'UUU',
 'imaging_depth': 150,
 'field_of_view_height': 512,
 'experiment_container_id': '1',
 'imaging_plane_group': 0,
 'mouse_id': 457841,
 'sex': 'F',
 'age_in_days': 233,
 'full_genotype': 'Sst-IRES-Cre',
 'reporter_line': 'Ail48(TIT2)',
 'driver_line': ['Sst-IRES-Cre'],
 'cre_line': 'Sst-IRES-Cre',
 'date_of_acquisition': '2015-08-11',
 'ophys_frame_rate': 11.0,
 'indicator': 'GCaMP6f',
 'targeted_structure': 'VISp',
 'excitation_lambda': 910.0,
 'emission_lambda': 520.0}
```

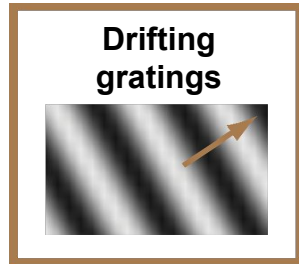
plot pupil area for the same experiment

```
fig, ax = plt.subplots(1,1, figsize = (20,5))
ax.plot(dataset.eye_tracking.timestamps, dataset.eye_tracking.pupil_area, color='gray')
ax.set_xlabel('time (seconds)')
ax.set_ylabel('pupil area (pixels^2)')
ax.set_title('Ophys experiment {}'.format(ophys_experiment_id), fontsize = 20)
```

```
Text(0.5, 1.0, 'Ophys experiment 957759562')
```



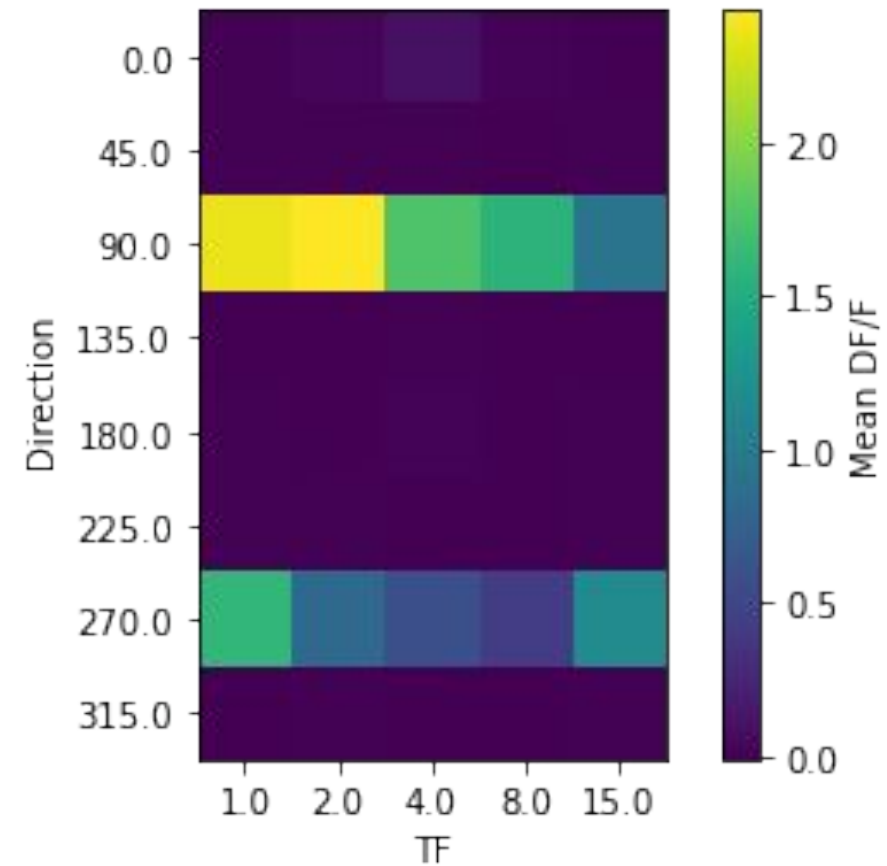
Today's tutorial:



- Max projection image
- ROI masks
- Cell specimen IDs
- Raw fluorescence for each ROI
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- Stimulus table

Experiment Session A

- Stimulus template (images/movies)
- Mouse's running speed
- Pupil position and area
- Motion correction (x & y pixels)
- Meta data (cre line, visual area, imaging depth, experiment container, mouse ID, age, sex, etc)



You will learn:

- Learn how to integrate neural activity and stimulus information from NWB files
- Compare tuning properties of different neurons of the same type. What kind of variability do you see?
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